

LEE SHING HEI, CYRUS

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Education

The Chinese University of Hong Kong

Expected May 2026

Bachelor of Engineering in Computer Engineering

- CUHK Faculty of Engineering Admission Scholarship
- Professor Charles K. Kao Student Creativity Award (PCKKSCA) 2025 - Undergraduate Group Champion

Experience

Hong Kong Centre for Logistics Robotics

May 2025 – Present

Engineer Intern

- Developed a high-performance, bilingual corporate website using Next.js 15 and Strapi CMS, featuring 3D interactions (Three.js) and Server-Side Rendering (SSR) to maximize SEO and brand engagement
- Optimized the deployment workflow by containerizing the full stack with Docker and implementing a CI/CD pipeline using GitHub Actions, ensuring automated, zero-downtime updates for the production site

Fase Technology Group Limited

Jun 2024 – Oct 2024

Robotics Engineer Intern

- Developed kinematic control algorithms for DOBOT and AUBO robotic arms using Python and C++
- Engineered a real-time computer vision pipeline on NVIDIA Jetson Nano using TensorRT, achieving over 90% accuracy in object detection
- Implemented TCP/IP socket communication between edge devices and microcontrollers for synchronized actuation

Peanut KING Solution Limited

Jun 2023 – May 2024

Software Engineer Intern

- Developed, optimized and deployed a production-grade educational platform using React.js 18, Django, Docker and AWS EC2, serving 10+ Hong Kong schools and educational institutions with over 200 students

Projects

Hierarchical Metric-Semantic Edge Fusion Navigation System | NVIDIA Jetson, ZeroMQ, TensorRT, OpenCV, ESP32

- Implemented a distributed microservices system on NVIDIA Jetson Orin Nano using ZeroMQ, decoupling real-time safety control (C++/Python) from high-latency AI services
- Optimized SegFormer and Moondream2 VLM using TensorRT (FP16) and llama.cpp 4-bit Quantization, achieving a 35% improvement in performance with 30 FPS inference on constrained edge hardware (8GB)
- Designed a custom Semantic Polar Histogram (SPH) algorithm for reactive path planning, replacing standard navigation stacks to reduce latency by 20%
- Implemented a hardware-level reflex loop using ESP32 and ultrasonic sensors to physically detect obstacles

StudySage - AI-Powered Learning Assistant | Python, LLMs, Google Cloud APIs, React.js, Flask

- Developed an asynchronous processing pipeline integrating Google Cloud APIs and LLMs for real-time interaction

Dungeon Escape - FPGA SoC Game | C++, UART/I2C protocols, FPGA

- Designed a Hardware-Software Co-design architecture on Zynq-7000 FPGA, integrating custom hardware IP blocks for sensor processing

Technical Skills

Core Languages: JavaScript/TypeScript, C++, Python, SQL, Bash

Robotics & AI: PyTorch, TensorFlow, TensorRT, YOLOv8, OpenCV, ROS (Fundamental), LLM/VLM Integration

Embedded System: NVIDIA Jetson (Orin/Nano), ESP32, Arduino, UART/I2C protocols, FPGA (Zynq-7000)

Software & Infrastructure: Docker, Linux (Ubuntu), ZeroMQ, React.js/Next.js, Django, Flask, PostgreSQL, Git, CI/CD

Extra-Curricular Activities

CUHK Robocon Team - Core Member

Sep 2024 – Present

- Implemented computer vision algorithms using YOLOv8 and Python on NVIDIA Jetson platforms for semi-autonomous competition robots, collaborating on sensor fusion strategies
- Collaborated as a member of programming team, contributing to prototyping, testing and fine-tuning competition robot's mechanism using C++, Raspberry Pi, Arduino Due and VESC Tool